

Research vision

MAIA investigates how a vertically integrated, open AI stack can connect language-model training, scientific knowledge retrieval and computational tools while remaining deployable on local hardware and HPC infrastructure.

Goal: build reproducible AI infrastructure that can support research, education and computational modeling without depending on a single proprietary cloud service.

WHY MAIA?

Scientific AI workflows are often fragmented across proprietary models, isolated RAG systems and cloud-only services. MAIA explores an integrated alternative:

- OPEN** Open-source components and reproducible pipelines
- LOCAL** On-premise inference and data control
- MULTILINGUAL** English • Portuguese • Spanish
- SCIENTIFIC** Designed to connect LLMs with computational tools

RESEARCH OBJECTIVES

- 1 Pretrain a compact multilingual foundation model from scratch.
- 2 Fine-tune domain models for computational modeling and academic tasks.
- 3 Integrate RAG, symbolic/computational tools and conversational interfaces.
- 4 Deploy the stack across local machines, edge nodes and HPC resources.

MAIA LITE: PRETRAINING

- 354.8M parameters
- GPT-like
- 50,257 vocab

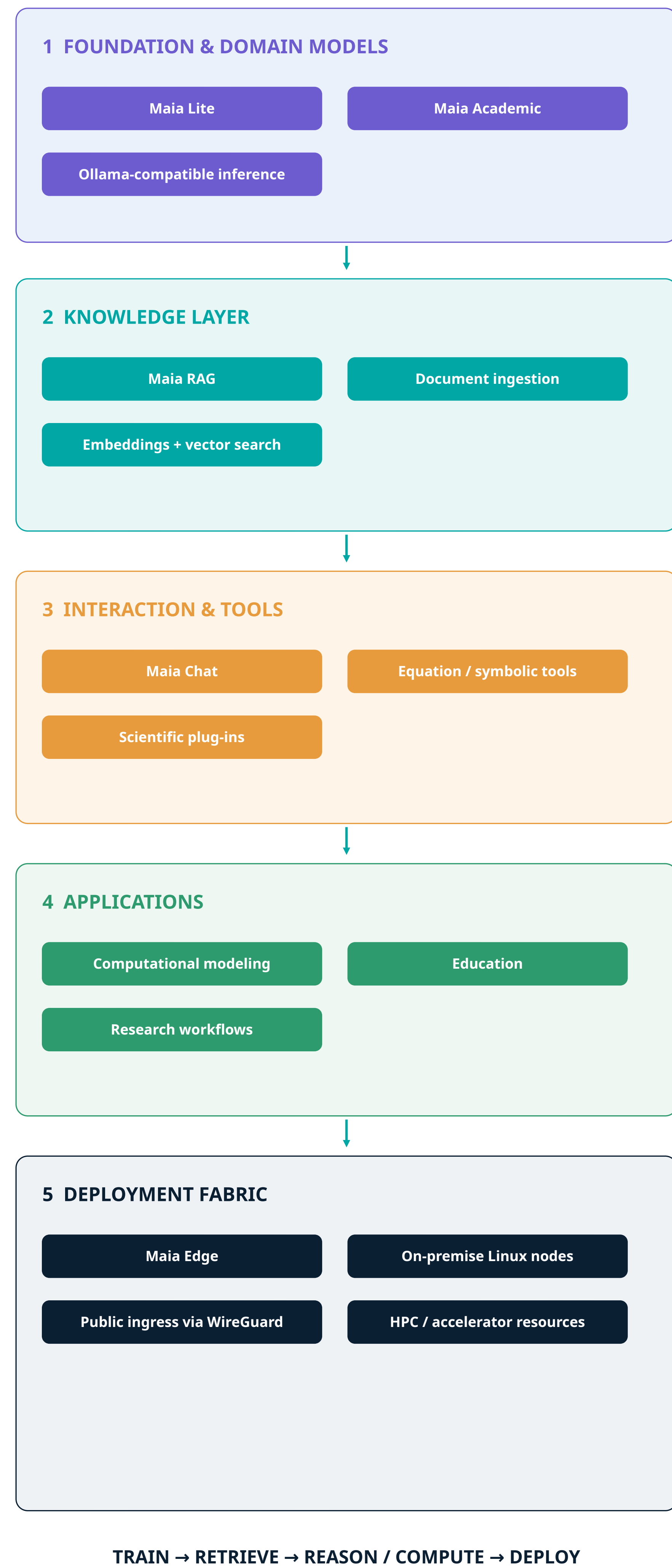
Training design

- Corpus** Wikipedia streams in EN / PT / ES
- Target** 300k optimization steps
- Compute** L4 experiments → A100 80 GB continuation
- Checkpointing** Periodic checkpoints with held-out evaluation
- Next phase** Unified fine-tuning: conversation + Python + computational modeling

Status: pretraining in progress; evaluation is checkpoint-based and separated from training data.

THE MAIA ECOSYSTEM

A layered architecture from model training to scientific applications



TRAIN → RETRIEVE → REASON / COMPUTE → DEPLOY

CURRENT IMPLEMENTATION

- Maia Chat** Web conversational interface backed by local LLM inference.
- Maia RAG** Automated ingestion architecture for PDFs, Office files and source code.
- Maia Edge** Public gateway + WireGuard tunnel to on-premise application nodes.
- Scientific tools** Planned bridge from natural language to symbolic and numerical computation.

DATA & FINE-TUNING PIPELINE

- 1,000 scientific papers
- Text extraction & chunking
- Local LLM Q&A generation
- Quality filtering
- EN → PT-BR / ES translation
- Unified supervised fine-tuning

EXPECTED CONTRIBUTIONS

- 01 A compact multilingual model trained under a documented pipeline.
- 02 A reusable RAG layer for scientific and educational corpora.
- 03 An open deployment architecture spanning edge, local and HPC resources.
- 04 A platform for experiments connecting LLMs to computational modeling tools.

FROM LANGUAGE MODELS TO COMPUTATIONAL SCIENCE

MAIA is designed as an experimental bridge between generative models and executable scientific workflows.

